

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: V

Name of the Course: Bigdata Visualization

Course Code: TCS571

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)																
(a)	Compare and contrast Data Visualization, Information Visualization, and Concept Visualization. Provide examples of each and discuss their applications.	CO1															
(b)	Describe the methodology for visualization design. Discuss the key steps involved in establishing the intent of a visualization project.	CO1															
(c)	Discuss how metaphor visualization enhances understanding in data communication. Provide an example of an effective metaphor-based visualization.	CO1															
Q2																	
(a)	Explain the importance of maintaining a strong editorial focus in data visualization. How does it influence audience understanding and message clarity?	CO2															
(b)	Explain how visual data analysis helps in identifying stories within datasets. Provide a case example of discovering a hidden insight through visualization.	CO2 CO2															
(c)	Explain the concept of interactivity, annotation, and arrangement in data presentation. How do these enhance user engagement and comprehension?	CO2															
Q3																	
(a)	The following dataset represents water distribution in a smart city (in million liters): Using this dataset: <table><tr><th>Source</th><th>Target</th><th>Flow (ML)</th></tr><tr><td>Reservoir</td><td>Treatment</td><td>60</td></tr><tr><td>River</td><td>Treatment</td><td>40</td></tr><tr><td>Treatment</td><td>Households</td><td>70</td></tr><tr><td>Treatment</td><td>Leakage</td><td>20</td></tr></table> 1. Draw a Sankey Diagram to visually represent the water flows. 2. Calculate the total water input to the treatment plant. 3. Identify which sector consumes the largest share of water. 4. Explain how the Sankey diagram helps in analyzing efficiency and losses in this water distribution system.	Source	Target	Flow (ML)	Reservoir	Treatment	60	River	Treatment	40	Treatment	Households	70	Treatment	Leakage	20	CO4 CO3 CO3
Source	Target	Flow (ML)															
Reservoir	Treatment	60															
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(b)	You are analyzing student performance metrics across multiple schools, including test scores, attendance rates, and extracurricular participation. How do you choose the most effective visualization depending on whether the data is quantitative, categorical, or temporal? Compare chart types with examples relevant to education data.																
(c)	Differentiate between scatter plot, bubble plot, and heatmap in representing																

	relationships between variables. Provide an example dataset for each and explain how insights differ.	
Q4		
(a)	Describe the ElastiCube technology used in Sisense. How does Sisense handle large-scale data for real-time visualization and business intelligence reporting?	CO5 CO5
(b)	Describe the working mechanism of Google Charts. Discuss how it utilizes web technologies for dynamic visualization and provide an example of how to create a dashboard using Google Charts API.	CO4
(c)	Compare and contrast Tableau, Sisense, and IBM Watson Analytics based on data integration capabilities, ease of use, and advanced analytics support.	
Q5		
(a)	Write the step-by-step process to create a dashboard in Tableau that connects to data and builds a basic visualization showing Sales for each Sub-Category, broken down by Order Date, Year versus Category.	CO5 CO6
(b)	Describe the steps involved in creating and customizing interactive dashboards in Tableau. Explain how filters, actions, and layout options contribute to interactivity and user engagement.	CO6
(c)	Explain the concept of aggregation, granularity, and level of detail (LOD) in Tableau. How do LOD expressions help in controlling data visualization accuracy?	